

Ex.N.3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation

Loss.

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AIM:To analyze the propagation loss characteristics of different frequency bands (Sub-1 GHz, Mid-band, and mmWave) in wireless communication systems and compare their performance over varying distances using MATLAB.

Objective :

To compute the Propagation Loss (dB) for Sub-1 GHz, Mid-band, and mmWave frequency ranges using standard path loss models.

Procedure:

1. Define frequency bands: Sub-1 GHz (0.9 GHz), Mid-band (3.5 GHz), and mmWave (28 GHz).
2. Set communication distance range (e.g., 0.1 km to 5 km).
3. Use Free Space Path Loss (FSPL) formula to compute propagation loss:
$$PL(\text{dB})=20\log_{10}(d)+20\log_{10}(f)+32.44$$
4. Calculate propagation loss for each frequency band over varying distances.
5. Plot distance vs propagation loss curves for comparison.
6. Analyze how higher frequencies experience higher attenuation.
7. Compare coverage efficiency of each spectrum band.
8. Display results using MATLAB plots.

Matlab Code :

```
clc; clear; close all;
```

```
d = 0.1:0.1:5; % Distance (km)
```

```
f = [900 3500 28000]; % Frequency (MHz)
```

```
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
```

```
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
```

```
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(d,PL1,'b','LineWidth',2); hold on
```

```
plot(d,PL2,'r','LineWidth',2)
```

```
plot(d,PL3,'k','LineWidth',2)
```

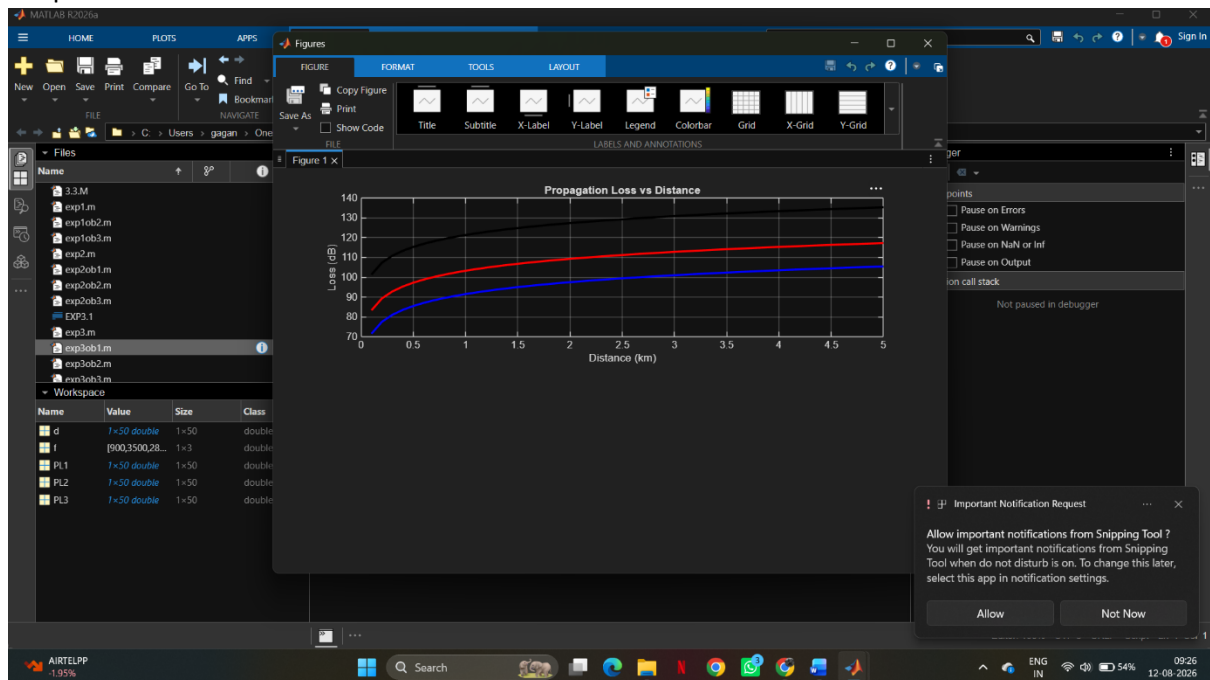
```
title('Propagation Loss vs Distance')
```

```
xlabel('Distance (km)')
```

```
ylabel('Loss (dB)')
```

```
grid on
```

Output:



Result:

1. The propagation loss was successfully computed for Sub-1 GHz, Mid-band, and mmWave frequency bands using the Free Space Path Loss (FSPL) model.
2. It was observed that propagation loss increases with increase in operating frequency, with mmWave showing the highest attenuation and Sub-1 GHz showing the lowest loss.
3. The comparative analysis of spectrum bands clearly demonstrates that lower frequency

bands are more suitable for long-range communication, while higher frequency bands require advanced techniques like beamforming for efficient performance.